

Requirement Analysis Phase

Date: 05 February 2025

Team Members

Mangalagiri Tejaswi — Team Lead

Putta Harshitha — Member

Deva Charmi Kondapalli — Member

Mohammad Sabir Baig — Member

Orchu Likhitha — Member

Project Name

Cab Booking System

Maximum Marks: **5 Marks**

1. Problem Statement

Many users face difficulties in booking cabs quickly, checking cab availability, and managing ride information. Traditional booking methods are time-consuming and lack proper tracking. Administrators also require an efficient system to manage users, bookings, and cab information from a centralized platform.

2. Functional Requirements

- User Registration
 - User Login
 - View Available Cabs
 - Book a Cab
 - Cancel Booking
 - View Booking History
 - User Profile Management
 - Admin Dashboard
 - Manage Users
 - Manage Cab Details
-

3. Non-Functional Requirements

- Secure Authentication
- Fast System Response
- Responsive User Interface
- High Availability
- Reliable Database
- Easy Navigation

- Data Security
 - Scalability
-

4. User Requirements

User

- Register and Login
- Search Available Cabs
- Book a Ride
- View Booking History
- Cancel Booking
- Update Profile

Admin

- Manage Users
 - Manage Cab Details
 - Monitor Bookings
 - Update Ride Information
-

5. Software Requirements

- Windows 10/11

- Node.js
 - React.js
 - Express.js
 - MongoDB
 - Visual Studio Code
 - Git
 - GitHub
 - Postman
 - Google Chrome
-

6. Hardware Requirements

- Intel Core i5 Processor or above
 - Minimum 8 GB RAM
 - 1 GB Free Storage
 - Internet Connection
 - Laptop/Desktop
-

7. Technology Stack

Frontend

- HTML
- CSS
- JavaScript
- React.js
- Bootstrap

Backend

- Node.js
- Express.js

Database

- MongoDB

8. Assumptions

- Users have internet connectivity.
- MongoDB server is available.
- Admin manages cab and user information.
- Users provide valid registration details.

9. Constraints

- Internet connection is required.

- System depends on MongoDB availability.
 - Authentication is mandatory before booking a cab.
-

10. Expected Outcome

The Cab Booking System will provide a secure, user-friendly, and efficient platform for booking cabs online. Users can easily register, log in, book rides, manage bookings, and view ride history, while administrators can efficiently manage users, cab details, and bookings through a centralized dashboard.